# Muhammad Daffa ' Alfiannoor

Informatics Engineering Student | Data Science & Machine Learning
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## SUMMARY

Informatics Engineering student at Gunadarma University focused on Data Science, Machine Learning, and AI. Experienced building end-to-end ML pipelines and full-stack applications, from data preprocessing and model development to deployment, spanning health-risk prediction, workforce competency analytics, banking lead scoring, e-commerce and retail analytics, and NLP-based sentiment analysis.

## SKILLS

**Languages:** Python, SQL

**Machine Learning & Deep Learning:** Scikit-learn, TensorFlow/Keras, XGBoost, NLP (IndoBERT, Sastrawi), RandomizedSearchCV, StratifiedKFold

**Generative AI:** RAG, LLM APIs (Claude), LangChain, Hugging Face (sentence-transformers), ChromaDB

**Data Analysis & Visualization:** Pandas, NumPy, EDA, Data Wrangling, Statistical Analysis (hypothesis testing, chi-square), RFM segmentation, K-Means clustering, CLV analysis, PowerBI

**Web Development:** FastAPI, React, Vite, Tailwind CSS

**Tools:** Git, Docker, SQLite, PostgreSQL

## PROJECTS

**VitaCheck - Health Risk Prediction System (Undergraduate Thesis)**

- Built a full-stack ML system (CRISP-DM) classifying users into Low/Medium/High health risk on a 15,000-row lifestyle dataset; compared Logistic Regression, Random Forest, XGBoost, and SVM.
- Selected SVM as the final model, prioritizing recall on the High Risk class (0.9313) over raw accuracy; overall accuracy 0.9180, F1 0.9178, weighted AUC-ROC 0.9892.
- Used permutation importance to identify sleep duration, physical activity, and pre-sleep phone usage as the top predictors.
- Backend: FastAPI, JWT authentication, SQLite. Frontend: React, Vite, Tailwind CSS, Recharts, including a prediction-history dashboard.

**RumahKarir - IT Skill Gap Detection System**

- AI-based system identifying employee competency gaps from performance, skills, experience, and career interest data, against SKKNI/SFIA standards, using NLP (IndoBERT, Sastrawi).
- Analyzed 123,849 job postings across 24,473 companies and 35 skill categories; redesigned the Gap Score methodology using IQR-based thresholding to address class imbalance.
- Found remote postings pay a median of US$113,488 vs US$78,000 for non-remote ($p < 0.0001$), and high-skill postings earn a US$6,754 premium over low-skill postings ($p < 0.0001$).

**Predictive Lead Scoring Portal for Banking Sales**

- Built a predictive system that analyzes prospective customer data to generate a lead score indicating likelihood of conversion, supporting bank sales prioritization.

**E-Commerce Data Analysis**

- Analyzed 99,441 orders and 112,650 order items to identify sales trends and customer behavior patterns; 97.0% of orders were delivered successfully.
- Identified health_beauty as the top-revenue category (R$1,233,132) and segmented customers via RFM, finding the "Cant Lose Them" segment drove 22.8% of total revenue (R$3,512,699.19).

**Sentiment Analysis on the Shopee App**

- Built a positive/negative sentiment classifier on 56,614 cleaned Shopee app reviews scraped from the Google Play Store.
- Achieved 85.18% accuracy with Logistic Regression + TF-IDF, outperforming Naive Bayes (75.31%), Random Forest + Word2Vec (76.05%), and Decision Tree (73.71% test accuracy, indicating overfitting).

**Online Retail Analysis - RFM Segmentation & Customer Lifetime Value**

- Cleaned a 541,909-row UK online retail transaction dataset down to 392,692 rows and performed RFM segmentation, K-Means clustering, and CLV analysis across 4,338 unique customers (total revenue GBP 8,911,407.90).
- Identified a 311-customer "Champions" cluster (silhouette score 0.66) with average monetary value of GBP 15,620.40, against an overall average CLV of GBP 2,048.69 and 79.4% monthly churn.

**Customer Churn Analysis & Prediction**

- Built an end-to-end churn prediction pipeline on the Telco Customer Churn dataset (7,043 customers, 19 features), covering data quality audit, EDA, statistical testing (chi-square, Cramér's V), and a leakage-free scikit-learn preprocessing pipeline.
- Compared five classifiers with 5-fold stratified cross-validation and RandomizedSearchCV; tuned Gradient Boosting reached a test ROC-AUC of about 0.843, with an F1-optimal decision threshold of 0.35 to handle the 26.5% churn imbalance.
- Found month-to-month contracts churn at 42.7% versus 2.8% for two-year contracts; used permutation importance and Low/Medium/High risk segmentation to help prioritize retention efforts.
- Delivered an interactive 7-tab Streamlit dashboard with KPIs, model performance, interpretability, and a real-time churn prediction form.

**Netflix Catalog Analysis - Recommendation, Forecasting & Classification**

- Built an end-to-end data science project on a Netflix catalog snapshot, cleaning 8,790 raw rows into 8,782 titles with 34 engineered features, followed by EDA (movies 69.7% of the catalog; US the largest producer at 38.1%) and an interactive Streamlit business dashboard.
- Developed a content-based recommender using TF-IDF and cosine similarity on genre, rating, type, and country metadata; forecasted catalog additions by comparing naive, linear, polynomial, Holt-Winters, and logistic growth models with rolling-origin backtesting.
- Classified titles as mature vs. family-oriented by comparing Logistic Regression, Random Forest, HistGradientBoosting, and LightGBM (about 0.84 test accuracy, 0.96 recall on mature titles); documented a target-leakage case in the original Movie vs. TV Show framing.

**Regulasi Q&A Assistant - Retrieval-Augmented Generation (RAG)**

- Built a RAG-based assistant for answering questions over Indonesian regulation PDFs, using PDF text extraction, Pasal-aware document segmentation, and recursive character chunking with LangChain.
- Generated multilingual embeddings with Hugging Face's multilingual-e5-small and implemented cosine-similarity retrieval using ChromaDB.
- Integrated an OpenAI-compatible LLM API to generate context-grounded Indonesian answers, with Pasal-level source display and a prompt-based "information not found" response for unsupported questions.
- Developed FastAPI and Streamlit interfaces, including extractive retrieval mode when the LLM API is unavailable, and containerized the application with Docker Compose.
- Implemented a configurable retrieval evaluation harness using Hit Rate@k and MRR@k to compare chunk sizes and top-k configurations.

## EXPERIENCE

**Machine Learning Cohort**

*ASAH led by Dicoding* *2025 - 2026*

- Built and deployed ML models for real-world use cases using Python and TensorFlow/Keras, covering the full lifecycle from data preparation to deployment.

**Data Scientist Cohort**

*Coding Camp powered by DBS Foundation* *2026*

- Practiced end-to-end data science workflows: preprocessing, EDA, feature engineering, model building and evaluation, and visualization through real-world case studies.
- Served as team leader for the capstone project.

## EDUCATION

**Bachelor's Degree, Informatics Engineering**

*Gunadarma University, Depok, Jawa Barat* *2023 - Present*